NON-MF GROUPS

SAUERS

Not every group is MF: $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ is a simple property-(T) group where every homomorphism from H to an MF group is trivial, so $C^*_{\mathrm{r}}(H)$ is a separable stably finite C^*-algebra that is not MF. In general, a normal property-(T) subgroup contained in the commutators of a one-sided compression of a property-(T) subgroup is killed by every homomorphism to an MF group. The same criterion gives a finitely presented torsion-free property-(T) group with no nontrivial homomorphism to an MF group.

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1. Introduction

The counterexample is

$$H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2)),$$

the elementary group of 12×12 matrices over the binary Leavitt algebra. Khanh–Thanh show that H is isomorphic to the unit group of the binary Leavitt algebra [K–T, Proposition 4.2 and Corollary 4.4], which OpenAI used to construct the first nonsofic group [OAI, Chapter 3].

A countable group G is *MF* [CDE] if there are positive integers d_n and maps $V_n\colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

In other words, G is MF if there are finite unitary matrices, one for each element of G, that multiply correctly up to an error tending to zero in operator norm while every element other than the identity stays a fixed distance from the identity matrix. The separation may equivalently be required along the full sequence with a constant independent of g [Kor, Propositions 2 and 7]. The strong convergence convention of [GKM, Sch] also requires the models to reproduce the operator norms of the left regular representation, so a

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group that is not MF as defined here is not MF in that convention either. Equivalently, G embeds in the unitary group of the norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}),$$

where $\prod_n M_{d_n}(\mathbb{C})$ is the C^*-algebra of bounded sequences, as every product of C^*-algebras below, and $\bigoplus_n M_{d_n}(\mathbb{C})$ is its ideal of sequences with $\|x_n\| \to 0$. Since $\prod_n M_{d_n}(\mathbb{C})$ is the multiplier algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$, the quotient $\mathcal{Q}_{\mathbf{d}}$ is its corona algebra, and we call a homomorphism $G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. A separable C^*-algebra is MF if it embeds in a norm matrix corona [BK]; for a countable group G, the algebra $C^*_{\mathrm{r}}(G)$ is separable and its faithful canonical trace makes it stably finite.

The groups here fail to be MF because they contain a subgroup that is conjugated into itself. For $L \le G$ put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \le L\}.$$

An element c of the centralizer $C_G(L)$ commutes with L, so ucu^{-1} commutes with uLu^{-1}; it need not commute with the rest of L. The *compression–centralizer defect* of L is the normal subgroup of G generated by the commutators of all such elements ucu^{-1} with elements of L,

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \le G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \le \mathfrak{D}_G(L)$, then every homomorphism from G to an MF group is trivial on K. In particular, a nontrivial such K makes G non-MF, and if G has property* (T) *and $\mathfrak{D}_G(L) = G$, then every homomorphism from G to an MF group is trivial.*

Asymptotic multiplicativity in operator norm makes the conjugation maps $\mathrm{Ad}(V_n(g))\colon x \mapsto V_n(g)xV_n(g)^*$ an asymptotic unitary representation of G on $M_{d_n}(\mathbb{C})$ with its normalized Hilbert–Schmidt inner product, because $\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \le 2\,\|A - B\|$. Property (T) applies to unitary representations on Hilbert spaces, so it applies to this one, and its conclusions are estimates in Hilbert–Schmidt norm. In an ultraproduct of these Hilbert spaces the maps $\mathrm{Ad}(V_n(g))$ become a unitary representation of G. Let P be the projection onto the vectors fixed by L, which is the Kazhdan projection of L, and let U be the unitary of the compressor u. If ξ is fixed by L, then so is $U^*\xi$, because $\ell \cdot U^*\xi = U^*\big((u\ell u^{-1}) \cdot \xi\big)$ for $\ell \in L$ and $u\ell u^{-1} \in L$ fixes ξ. So $U^*PU \le P$. The projections P and U^*PU are conjugate by the unitary U, and in a stably finite C^*-algebra two equivalent projections, one below the other, are equal; the matrix corona is stably finite, so $U^*PU = P$. In terms of sequences, the vectors fixed by L are the classes of (x_n) with $\|[V_n(\ell), x_n]\|_2 \to 0$ for all $\ell \in L$, and $U^*PU = P$ means that conjugation by $V_n(u)$ maps this Hilbert–Schmidt asymptotic commutant of L onto itself. For $c \in C_G(L)$ the sequence $(V_n(c))$ lies in the asymptotic commutant, so $(V_n(ucu^{-1}))$ does too, that is, $\big\|V_n([ucu^{-1}, \ell]) - 1\big\|_2 \to 0$ for all $\ell \in L$. The elements with this property form a normal subgroup, so every element of $\mathfrak{D}_G(L)$ tends to 1 in Hilbert–Schmidt norm.

So the hypothesis $K \le \mathfrak{D}_G(L)$ gives $\|V_n(k) - 1\|_2 \to 0$ for $k \in K$, and a second use of property (T), now for K, gives triviality in operator norm. Suppose that a corona homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Let $p \in \mathcal{Q}_{\mathbf{d}}$ be the Kazhdan projection of K, the projection onto the K-fixed vectors. Then $p \neq 1$, and p commutes with $\Theta(G)$ because K is normal. So Θ compresses to the corner $(1-p)\mathcal{Q}_{\mathbf{d}}(1-p)$, where K has no fixed vectors, and the compressed homomorphism lifts to an operator norm asymptotic representation (W_n) of G on smaller matrices. The Kazhdan inequality for K bounds the average of $\|W_n(s) - 1\|_2^2$ over a Kazhdan set $S \subseteq K$ below by a positive constant, so some $s \in S$ stays a fixed Hilbert–Schmidt distance from 1, against the vanishing on K applied to (W_n). So every corona homomorphism is trivial on K, and by Lemma 2.1 so is every homomorphism to an MF group. A Hilbert–Schmidt bound on the multiplicative defect gives no operator norm bound on the conjugation maps, so the argument does not apply to sofic or hyperlinear approximations.

Compressors are easy to find: the stable letter of an ascending HNN extension of a property-(T) group along a proper self-embedding is one. The group H realizes the compression inside a simple group, where a single nontrivial defect element normally generates everything. Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra, the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group). *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is finitely generated, nontrivial, simple, has property* (T)*, and every homomorphism from H to an MF group is trivial. So H is not MF, and $C^*_{\mathrm{r}}(H)$ is separable and stably finite but not MF.*

Theorem C (a torsion-free finitely presented example)**.** *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property* (T) *such that every homomorphism from Q to an MF group is trivial. In particular, no nontrivial quotient of Q is MF.*

Section 4 builds Q from Fournier-Facio's torsion-free property-(T) group [FF, §2] and Hull's small cancellation theorem [Hu]; it uses a different toolkit from the rest of the paper and can be read independently.

Blackadar and Kirchberg introduced MF algebras and proved that a separable C^*-algebra is NF if and only if it is nuclear and MF [BK]; Carrión–Dadarlat–Eckhardt introduced the group property [CDE]. The negative solution of the Connes embedding problem [JNVWY] implies the existence of separable stably finite C^*-algebras that are not MF [GH, Proposition 6.1 and Remark 6.2]; Theorem B gives one among reduced group C^*-algebras.

Nonapproximability was known for the unnormalized Schatten p-norms when $1 < p < \infty$ [DGLT, LO], and Bachner–Dogon–Lubotzky settled $p = 1$ [BDL]; the case $p = \infty$ remained open. For groups of Deligne type, operator–Hilbert–Schmidt stability would imply that the group is not MF [BDL, Proposition 1.5]; here the input is the compression relation instead.

2. One-sided compression

2.1. **Corona homomorphisms.** The image of a corona homomorphism from G is a countable subgroup of a corona unitary group, so MF, and every countable MF group embeds in the unitary group of a norm matrix corona.

Lemma 2.1. *Let G be countable and $K \leq G$. Every homomorphism from G to an MF group is trivial on K if and only if every corona homomorphism from G is trivial on K.*

Proof. A corona homomorphism is a homomorphism to an MF group, and a homomorphism to an MF group composed with an embedding of that group into a corona unitary group is a corona homomorphism with the same kernel. □

2.2. **Kazhdan transport in normalized Hilbert–Schmidt norm.** For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n \colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

The elements g with $\|V_n(g) - 1\|_2 \to 0$ form a normal subgroup of G: the operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, and this norm is unitarily invariant, so $\|V_n(gh) - 1\|_2 \leq \|V_n(g) - 1\|_2 + \|V_n(h) - 1\|_2 + o(1)$ and $\|V_n(ghg^{-1}) - 1\|_2 = \|V_n(h) - 1\|_2 + o(1)$. For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.2. *For every sequence of positive integers (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) \Big/ \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Write A for the displayed corona. If $v^*v = 1$ in A and (x_n) is a bounded lift of v, then $\|x_n^*x_n - 1\| \to 0$, so for all large n the matrix $u_n = x_n(x_n^*x_n)^{-1/2}$ is unitary and $\|u_n - x_n\| \to 0$; after assigning arbitrary unitary values to the finitely many remaining coordinates, (u_n) is a unitary lift of v, and $vv^* = 1$. So A is finite, and so is $M_k(A)$, which is the corona with matrix sizes km_n. Finally, if $p \leq q$ are equivalent projections in a stably finite algebra B and w is a partial isometry with $w^*w = q$ and $ww^* = p$, then $w \in qBq$ is an isometry in the finite corner qBq, so $p = ww^* = q$. □

Lemma 2.3 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$; then P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If ξ is L-fixed and $\ell \in L$, then $\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi$ because $U\pi(\ell)U^*$ belongs to $\pi(L)$. So $U^* \operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$, and the range projection U^*PU is dominated by P in $B(\mathcal{H})$, and then in B by faithfulness. □

Theorem 2.4 (one-sided Kazhdan transport)**.** *Let $L \leq G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\operatorname{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and a free ultrafilter ω. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Since

$$\|\operatorname{Ad}(A) - \operatorname{Ad}(B)\| \leq 2\,\|A - B\|$$

for unitaries A, B, the conjugation operators $\operatorname{Ad}(V_n(g))$ are multiplicative modulo c_0, so they define a unitary representation σ of G on $\mathcal{K}_\omega$, and the class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\operatorname{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})),$$

which is a norm matrix corona with coordinate sizes d_n^2 after choosing matrix units on each $M_{d_n}(\mathbb{C})$. By the estimate above, $g \mapsto \widetilde{\sigma}(g)$ is a homomorphism $G \to \mathcal{U}(\mathcal{B})$, so its restriction to L extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]; under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\operatorname{Fix}\sigma(L)$. Since $uLu^{-1} \leq L$, the unitary

$U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 2.3, so $U^*PU \le P$ inside $\mathcal{B}$. The two projections are unitarily equivalent, so Lemma 2.2 gives $U^*PU = P$: U commutes with P, and both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence: a subsequence staying above some $\varepsilon > 0$ would lie in a free ultrafilter and give a nonzero ultralimit. So $\mathrm{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$. □

Corollary 2.5. *Let $L \le G$ have property* (T) *and let (V_n) be an operator norm asymptotic representation of G. Then*

$$\|V_n(d) - 1\|_2 \longrightarrow 0 \qquad (d \in \mathfrak{D}_G(L)).$$

Proof. Let $u \in \mathrm{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$. Since c commutes with L, the sequence $(V_n(c))$ lies in $\mathcal{C}_2(V, L)$, and by Theorem 2.4 so does $(V_n(u)V_n(c)V_n(u)^*)$, which asymptotic multiplicativity identifies in operator norm with $(V_n(ucu^{-1}))$. So $\big\|[V_n(\ell), V_n(ucu^{-1})]\big\|_2 \to 0$, that is, $\big\|V_n([ucu^{-1}, \ell]) - 1\big\|_2 \to 0$. The elements with this property form a normal subgroup of G, so it contains $\mathfrak{D}_G(L)$. □

2.3. **From Hilbert–Schmidt to operator norm.** A corona homomorphism ρ cannot simply be restricted to a corner: a projection q commuting with $\rho(G)$ has lifts q_n that only asymptotically commute with the lifts of $\rho(g)$, so the compressions $q_n\rho(g)q_n$ are only approximately unitary. The next lemma corrects them.

Lemma 2.6 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \mathrm{rank}(q_n)$, unitary identifications $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$ such that, in the corona over the retained coordinates, the class of $(J_nW_n(g)J_n^*)$ is the coordinate restriction of $q\rho(g)$, a unitary of the corner $q\mathcal{Q}_{\mathbf{d}}q$ with unit q.

Proof. Lift q to projections q_n by spectral rounding, and lift each $\rho(g)$ to unitaries $U_n(g)$ by polar correction, with $U_n(1) = I_{d_n}$. The commutation relation in the corona gives $\|q_nU_n(g) - U_n(g)q_n\| \to 0$ for each g. So $q_nU_n(g)q_n$, viewed on $q_n\mathbb{C}^{d_n}$, has both unitarity defects converging to zero, and for $g, h \in G$,

$$\begin{aligned}&\|q_nU_n(gh)q_n - q_nU_n(g)q_n\, q_nU_n(h)q_n\| \\ &\qquad \le \|U_n(gh) - U_n(g)U_n(h)\| + \|[q_n, U_n(h)]\| \longrightarrow 0,\end{aligned}$$

because $q_nU_n(g)U_n(h)q_n - q_nU_n(g)q_nU_n(h)q_n = q_nU_n(g)[U_n(h), q_n]q_n$ (compare [BDL, Proposition 2.4]). For each fixed g, let $\widetilde{W}_n(g)$ be the polar correction of $q_nU_n(g)q_n$ in the corner whenever both defects are at most $1/2$, and q_n at the finitely many earlier stages; then $\widetilde{W}_n(g)$ is unitary in the corner, $\left\|\widetilde{W}_n(g) - q_nU_n(g)q_n\right\| \to 0$, and $\widetilde{W}_n(1) = q_n$. Since $q \neq 0$, infinitely many q_n are nonzero; retain those coordinates, choose unitaries $J_n \colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and put $W_n(g) = J_n^*\widetilde{W}_n(g)J_n$. Conjugation by J_n preserves operator norms and multiplicative defects, so (W_n) is an operator norm asymptotic representation, and the estimates identify the class of $(J_nW_n(g)J_n^*)$ with the coordinate restriction of $q\rho(g)$. □

Theorem 2.7 (normal Kazhdan subgroups)**.** *Let G be countable and let $K \trianglelefteq G$ have property* (T)*. If every operator norm asymptotic representation (V_n) of G satisfies $\|V_n(k) - 1\|_2 \to 0$ for all $k \in K$, then every corona homomorphism from G is trivial on K.*

Proof. Suppose that a corona homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Replace G by $\Theta(G)$ and K by $\Theta(K)$: the new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients, and it still satisfies the hypothesis, because every operator norm asymptotic representation of $\Theta(G)$ pulls back along $G \to \Theta(G)$ to one of G.

Let $p \in \mathcal{Q}_{\mathbf{d}}$ be the image of the Kazhdan projection of K; under any faithful representation of the corona, p is the projection onto the K-fixed vectors. Normality of K gives $\Theta(g)p\Theta(g)^* = p$ for all g, because $gKg^{-1} = K$. Put $q = 1 - p$; it is nonzero, since $p = 1$ would make every element of K act trivially. Lemma 2.6 gives an operator norm asymptotic representation $W_n\colon G \to \mathcal{U}(r_n)$ whose corona homomorphism $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$. Since q commutes with $\Theta(K)$, the Kazhdan projection of K under $\widehat{\Theta}$ is the coordinate restriction of $qp = 0$; so $\widehat{\Theta}|_K$ has no nonzero fixed vectors, and for a finite Kazhdan set $S \subseteq K$ with constant $\kappa > 0$ the Kazhdan inequality gives

$$b = \frac{1}{|S|}\sum_{s\in S}(\widehat{\Theta}(s) - 1)^*(\widehat{\Theta}(s) - 1) \;\geq\; \frac{\kappa^2}{|S|}\,1$$

in $\mathcal{Q}_{\mathbf{r}}$. The coordinates $b_n = \frac{1}{|S|}\sum_{s\in S}(W_n(s) - I_{r_n})^*(W_n(s) - I_{r_n})$ represent b, so the negative part of $b_n - (\kappa^2/|S|)I_{r_n}$ tends to zero in operator norm, and taking normalized traces on $M_{r_n}(\mathbb{C})$ gives

$$\frac{1}{|S|}\sum_{s\in S}\|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

After passing to a subsequence, one fixed $s_0 \in S$ stays a positive Hilbert–Schmidt distance from I_{r_n}, against the hypothesis applied to (W_n) and $s_0 \in K$. □

Proof of Theorem A. By Corollary 2.5, every operator norm asymptotic representation of G satisfies $\|V_n(k) - 1\|_2 \to 0$ for $k \in K \leq \mathfrak{D}_G(L)$. Theorem 2.7 then makes every corona homomorphism trivial on K, and Lemma 2.1

makes every homomorphism to an MF group trivial on K. If $K \neq 1$, then G does not embed in an MF group, so G is not MF; and $K = G = \mathfrak{D}_G(L)$ gives the last assertion. $\square$

3. The binary Leavitt group

The Leavitt algebra $R = L_{\mathbb{F}_2}(1,2)$ is built so that one copy of the free module R is isomorphic to two copies: the relations (2) say that s_0, s_1 embed two copies of R side by side and t_0, t_1 read them back. So a matrix A over R can be copied into one of the two halves, with the identity on the other half, and this compression is an injective endomorphism of the matrix ring that fixes the identity. Inside a large enough elementary group the compression is realized by conjugation, which gives a compressor τ for a copy L of $\mathrm{EL}_3(R)$; one commutator computation gives a nontrivial element of $\mathfrak{D}_H(L)$, and simplicity of H turns it into all of H.

Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1,$$

so that $p + q = 1$ and $t_1 q s_1 = 1$; in particular $q \neq 0$. We use indices $0, \ldots, 11$ for 12×12 matrices and identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ of matrices $\operatorname{diag}(A, I_9)$. Define $\Psi\colon M_3(R) \to M_3(R)$ by

$$\Psi(A) = qI_3 + s_0 A t_0, \tag{3}$$

where $s_0 A t_0$ is the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q^2 = q$, $qs_0 = t_0 q = 0$, and $t_0 s_0 = 1$ show that Ψ is multiplicative, $\Psi(I_3) = (p+q)I_3 = I_3$, and $t_0 \Psi(A) s_0 = A$; so Ψ is an injective unital ring endomorphism and restricts to an injective group homomorphism $\mathrm{GL}_3(R) \to \mathrm{GL}_3(R)$. It is realized by conjugation as follows. Put

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix};$$

block multiplication with (2) gives $Y = X^{-1}$ and $X \operatorname{diag}(A, I_3) Y = \operatorname{diag}(\Psi(A), I_3)$. The matrix X need not lie in an elementary group, which is why there are twelve coordinates: with

$$\tau = \operatorname{diag}(X, Y) \in \mathrm{GL}_{12}(R) \tag{4}$$

one has

$$\tau \operatorname{diag}(A, I_9)\, \tau^{-1} = \operatorname{diag}(\Psi(A), I_9). \tag{5}$$

Lemma 3.1. *The matrix τ defined in (4) belongs to $\mathrm{EL}_{12}(R)$.*

Proof. By the Whitehead lemma [Mil, §3], $\operatorname{diag}(X, X^{-1})$ is a product of block-triangular matrices $\left(\begin{smallmatrix} I & B \\ 0 & I \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} I & 0 \\ B & I \end{smallmatrix}\right)$ with $B \in M_6(R)$, and each of these is the product of the elementary matrices $e_{r,6+s}(b_{rs})$, respectively $e_{6+r,s}(b_{rs})$, since distinct off-diagonal matrix units have product zero. $\square$

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \leq H$ denote the image of the embedding $A \mapsto \operatorname{diag}(A, I_9)$.

Proposition 3.2. *The groups H and L have property* (T)*, and $\tau \in H$ satisfies*

$$\tau L \tau^{-1} \leq L. \tag{6}$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring, so the theorem of Ershov and Jaikin-Zapirain gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$. For $i \neq j$ and $a \in R$, equations (5) and (3) give $\tau e_{ij}(a)\tau^{-1} = e_{ij}(s_0 a t_0) \in L$, and these elementary matrices generate L. □

Proposition 3.3. *The group $H = \mathrm{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], so it is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $N \leq \mathrm{GL}_{12}(R)$ is normalized by $\mathrm{EL}_{12}(R)$, Preusser's sandwich theorem [Pre, Theorem 3] gives an ideal $I \trianglelefteq R$ with $\mathrm{EL}_{12}(R, I) \leq N \leq C_{12}(R, I)$. Simplicity of R gives $I = 0$ or $I = R$; if $I = R$ then $N = H$. If $I = 0$, then N centralizes every $e_{ij}(a)$: centralizing $e_{ij}(1)$ for all $i \neq j$ forces $g = \lambda I_{12}$ with $\lambda \in R^\times$, and centralizing $e_{ij}(a)$ forces $\lambda \in Z(R)^\times$. Since $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], $N = 1$. Finally, $e_{01}(1) \neq 1$. □

Proposition 3.4. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c \tau^{-1}, \ell]$$

in H. Then

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad \langle\langle d \rangle\rangle_H = H.$$

Proof. The Steinberg relations give $[e_{ij}(a), e_{34}(1)] = 1$ for $i, j \in \{0, 1, 2\}$, and these matrices generate L; so $c \in C_H(L)$. Direct multiplication of the matrices in (4) gives

$$\tau c \tau^{-1} = e_{01}(q) e_{34}(1).$$

The second factor commutes with ℓ, and $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Since H is simple, d normally generates H. □

Proof of Theorem B. Proposition 3.2 gives property (T), which implies finite generation [BHV, Theorem 1.3.1]. The containment (6), the centrality of c relative to L, and Proposition 3.4 put d in $\mathfrak{D}_H(L)$, and $\langle\langle d \rangle\rangle_H = H$ gives $\mathfrak{D}_H(L) = H$. So Theorem A, applied with $K = H$, makes every homomorphism from H to an MF group trivial, and Proposition 3.3 makes H nontrivial and simple. In particular H is not MF. The algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite; if an embedding $e\colon C^*_{\mathrm{r}}(H) \to \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1 - p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$. □

4. A torsion-free finitely presented example

For an acylindrically hyperbolic group G, choose the generating set A provided by Hull [Hu, Theorem 3.12]. A subgroup is *suitable* if it acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup [Hu, Definition 1.4]. Hull's small cancellation theorem [Hu, Theorem 7.1] is stated with injectivity on a ball of $\Gamma(G, A)$; a ball containing a given finite set gives the following form.

Theorem 4.1 (Hull's small cancellation theorem)**.** *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A, and let $g_1, \dots, g_m \in G$ and a finite subset $\Omega \subseteq G$ be given. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is acylindrically hyperbolic, $\varphi|_\Omega$ is injective, $\varphi(g_i) \in \varphi(N)$ for all i, and every element of finite order in Q is the image of an element of the same order in G.*

Hull's proof treats $m = 1$ by passing to $G/\langle\langle W \rangle\rangle_G$ for one word W and the general case by induction on m [Hu, proof of Theorem 7.1], so $\ker \varphi$ is the normal closure of m elements and Q is finitely presented when G is.

Lemma 4.2 (saturation)**.** *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $\Omega \subseteq G$ be finite. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi|_\Omega$ is injective, and $\varphi(N) = Q$.*

Proof. The subgroup N is infinite and normal, so it acts non-elementarily on $\Gamma(G, A)$ by Osin [Os, Lemma 7.1], and torsion-freeness makes it suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable. Apply Theorem 4.1 to N_0 with $g_1, \dots, g_m$ a finite generating set of G and with the set Ω. Then $Q = \langle \varphi(g_1), \dots, \varphi(g_m) \rangle \leq \varphi(N_0) \leq \varphi(N) \leq Q$, so $Q = \varphi(N_0) = \langle \varphi(h_1), \varphi(h_2) \rangle$ and $\varphi(N) = Q$. The quotient is torsion-free by Theorem 4.1 and finitely presented because $\ker \varphi$ is normally generated by m elements. $\square$

Following Fournier-Facio [FF, §2], let U be a universal finitely presented torsion-free group, one containing a copy of every finitely presented torsion-free group [Chi]; let $F = F(a, b)$ be the free group of rank two; and let B_0 be a torsion-free hyperbolic group with property (T), obtained from the density model at a parameter between $1/3$ and $1/2$ [KK, Ol]. Small cancellation over the relatively hyperbolic pair $(U * B_0, U)$ gives a quotient Δ of $U * B_0$ in which U embeds and each generator of U equals an element of the image of B_0; so Δ is a finitely presented torsion-free quotient of B_0 [FF2, Proposition 2.3], torsion-freeness being preserved by Osin's embedding theorem [OsSC, Theorem 2.4(5)], and Δ has property (T). By universality, Δ contains a subgroup $\Delta_1 \times \Delta_2 \times F$ with $\Delta_i \cong \Delta$. Fix isomorphisms $\phi_i\colon \Delta \to \Delta_i$ and let E be the double HNN extension of Δ with stable letters u_1, u_2 and relations $u_i p u_i^{-1} = \phi_i(p)$ for $p \in \Delta$, so that $u_i \Delta u_i^{-1} = \Delta_i$. It is finitely presented, and torsion-free because an element of finite order fixes a vertex of the Bass–Serre tree and so is conjugate into Δ. The action of E on that tree is minimal and fixes no end, since Δ_1 and Δ_2 are proper subgroups of Δ [FF, §2], and the pointwise stabilizer of the vertices $u_1\Delta$ and $u_2\Delta$ is $\Delta_1 \cap \Delta_2 = 1$; so E is acylindrically hyperbolic by Minasyan–Osin [MO, Theorem 2.1]. Hull's common quotient theorem [Hu, Corollary 7.4], applied to the finitely generated torsion-free acylindrically hyperbolic groups E and B_0, gives an acylindrically hyperbolic group G_0 with surjections $\pi\colon E \to G_0$ and $B_0 \to G_0$, where π is injective on a prescribed finite subset of E because E has no nontrivial finite normal subgroup. In the proof of that corollary, G_0 is obtained from the free product $E * B_0$ by two applications of Theorem 4.1, each with finitely many elements g_i; so G_0 is torsion-free and, by the remark after Theorem 4.1, finitely presented. It has property (T) as a quotient of B_0. Put $w = [a, b] \neq 1$ in F and require the prescribed finite subset to contain 1 and w, so that $\pi(w) \neq 1$.

Put $\Gamma = \pi(\Delta)$, $t = \pi(u_1)$, and $J = t^{-1}\pi(F)t$. Since F commutes with $\Delta_1 = u_1\Delta u_1^{-1}$, the subgroup J centralizes Γ, and $t\Gamma t^{-1} = \pi(\Delta_1) \leq \Gamma$, so $t \in \mathrm{Comp}_{G_0}(\Gamma)$. Moreover $tJt^{-1} = \pi(F) \leq \Gamma$, and Γ has property (T) as a quotient of Δ.

Lemma 4.3. *For every homomorphism $\rho\colon G_0 \to \bar{G}$,*

$$\rho(\pi([F, F])) \leq \mathfrak{D}_{\bar{G}}(\rho(\Gamma)).$$

Proof. Write $\bar{\Gamma} = \rho(\Gamma)$, $\bar{t} = \rho(t)$, and $\bar{J} = \rho(J)$. Then $\bar{t}\bar{\Gamma}\bar{t}^{-1} \leq \bar{\Gamma}$, $\bar{J}$ centralizes $\bar{\Gamma}$, and $\bar{t}\bar{J}\bar{t}^{-1} = \rho(\pi(F)) \leq \bar{\Gamma}$. For $c \in \bar{J}$ and $\ell \in \bar{\Gamma}$, the commutator $[\bar{t}c\bar{t}^{-1}, \ell]$ lies in $\mathfrak{D}_{\bar{G}}(\bar{\Gamma})$ by (1). As c ranges over $\bar{J}$, the element $\bar{t}c\bar{t}^{-1}$ ranges over $\rho(\pi(F))$, and ℓ may be taken in $\rho(\pi(F)) \leq \bar{\Gamma}$. So the image of $[F, F]$ lies in $\mathfrak{D}_{\bar{G}}(\bar{\Gamma})$. □

Proof of Theorem C. Let N be the normal closure of $\pi([F, F])$ in G_0; it contains $\pi(w) \neq 1$. Lemma 4.2 applied to G_0, N, and the finite set $\{1, \pi(w)\}$ gives $\varphi\colon G_0 \to Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi(N) = Q$, and $\varphi(\pi(w)) \neq 1$. The group Q has property (T) as a quotient of G_0.

Let $r\colon Q \to M$ be a homomorphism to an MF group with image $\bar{G}$, and put $\rho = r \circ \varphi$. Since $\varphi(N) = Q$, the normal closure of $\rho(\pi([F, F]))$ in $\bar{G}$ is $\rho(N) = \bar{G}$, and by Lemma 4.3 it lies in the normal subgroup $\mathfrak{D}_{\bar{G}}(\rho(\Gamma))$; so $\mathfrak{D}_{\bar{G}}(\rho(\Gamma)) = \bar{G}$. The group $\bar{G}$ has property (T) as a quotient of Q, and $\rho(\Gamma)$ has property (T) as a quotient of Γ. Theorem A, applied to $\bar{G}$ with the subgroup $\rho(\Gamma)$ and $K = \bar{G}$, makes every homomorphism from $\bar{G}$ to an MF group trivial. But $\bar{G}$ is a subgroup of the MF group M, so it is MF, and its identity map is such a homomorphism; so $\bar{G} = 1$, and r is trivial. A homomorphism from a quotient of Q composes to one from Q, so no nontrivial quotient of Q is MF. □

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